1.

(a) It is sufficient to show $U \cap W = \{0\}$ and $\mathbb{R}^4 = U + W$.

$$U = \text{span}(u_1, u_2) = \{a_1 u_1 + a_2 u_2 \;,\; a_1, a_2 \in \mathbb{R}\}$$

$$= \{(a_1, a_1 + a_2, a_2, 0) \;,\; a_1, a_2 \in \mathbb{R}\}$$

similarly, $W = \{(b_2, 0, b_1, b_1 - b_2) \;,\; b_1, b_2 \in \mathbb{R}\}$

$v \in U \cap W \Rightarrow \exists\ a_1, a_2, b_1, b_2$ s.t $a_1 = b_2 = b_1 = a_2 = -a_1$

$\Rightarrow a_1 = a_2 = b_1 = b_2 = 0$

and $v = 0$.

$$U+W = \left\{ (a_1+b_2,\ a_1+a_2,\ a_2+b_1,\ b_1-b_2),\ \begin{matrix} a_1, a_2 \\ b_1, b_2 \end{matrix} \in \mathbb{R} \right\}$$

WTS $\dim U+W = 4$, as $U+W \subseteq \mathbb{R}^4$.

$(1,1,0,0)$ $(1,0,0,-1)$ $(0,1,1,0)$ $(0,0,1,1)$

u_1 w_2 u_2 w_1

These vectors clearly span $U+W$ as u_1, u_2 is a basis of U

and w_1, w_2 is a basis of W.

And they are also LI as none of the vectors is in the span of the others. $\therefore$ u_1, u_2, w_1, w_2 is a basis of $U+W \implies U \oplus W = \mathbb{R}^4$.

(b)

$b_1 = b_2$

$a_1 + a_2 = 1$

$a_2 + b_1 = 3$

$a_1 + b_2 = 2$

$2b_1 = 4$

$b_1 = 2 = b_2$

$a_1 = 0$

$a_2 = 1$

$(0+2, 0+1, 1+2, 2-2)$

$(0, 1, 1, 0)$

$+$

$(2, 0, 2, 0)$

2. V, W FDVS. $T: V \to W$ injection.

(a) $a_1 T(v_1) + a_2 T(v_2) + \cdots + a_n T(v_n) = 0$.

$\Rightarrow T(a_1 v_1) + \cdots + T(a_n v_n) = 0 = T(0)$

$\Rightarrow T(a_1 v_1 + \cdots + a_n v_n) = T(0)$

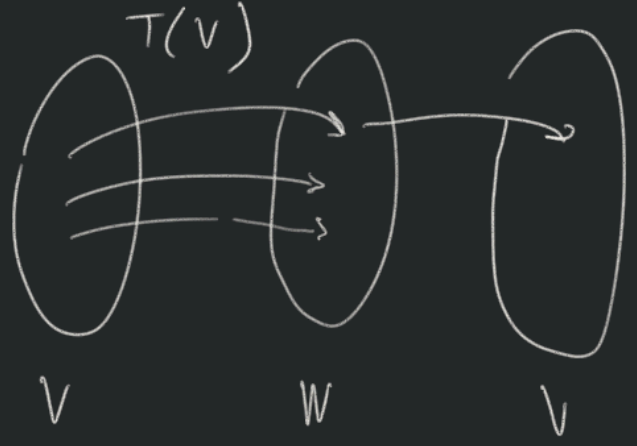
$\Rightarrow$ Because T injective we have

$a_1 v_1 + \cdots + a_n v_n = 0 \Rightarrow a_1 = \cdots = a_n = 0$

as $v_1 \cdots v_n$ is a basis.

(b) To show $\exists \; S : W \to V$ s.t. $ST = I_V$

We show that $\forall \; v \in V, \quad ST(v) = v$.



well-defined.



Because T is injective, $S(T(v)) = v$ suffices.

as $\quad T(a) = T(b) \Rightarrow a = b \quad \therefore \quad x = y \Rightarrow S(x) = S(y)$.

3. $T : \mathbb{R}^3 \to P_2(\mathbb{R})$

$T(a,b,c) = (a+b) + (b+c)x + (a+c)x^2$

basis $B = ((1,0,0), (1,1,0), (1,1,1))$ of $\mathbb{R}^3$

$C = (1, x, x^2)$ of $P_2(\mathbb{R})$

(a) Find $M(T)$ w.r.t. $\mathcal{B}$ and C.

$$\left.\begin{aligned} T(1,0,0) &= 1 \\ T(1,1,0) &= x \\ T(1,1,1) &= x^2 \end{aligned}\right\} \quad T(a,b,c) = (cx^2 - cx) + (bx - b) + a$$

$$M(T) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) Prove T is an isomorphism.

ISTS $M(T)$ is invertible, as $M(T) = I_3$, this is trivial.

4. Let V be finite-dimensional, $T: V \to W \in \mathcal{L}(V, W)$, $U \subseteq W$

Define $T^{-1}(U) = \{v \in V : Tv \in U\}$

Consider $S: T^{-1}(U) \to U \cap \text{range } T$ $\quad S(v) = Tv$.

(a) $\forall\, v \in T^{-1}(U)$, $\quad Tv \in U \implies S(v) \in U$.

$S(v) = Tv \implies S(v) \in \text{range } T$.

$\Big\}$ $S(v) \in U \cap \text{range } T$

Linear: $S(v)$ linear because T is linear.

Surjective $\forall\ u \in U \cap \text{range } T$:

$u \in \text{range } T \Rightarrow \exists$ some $v \in V$ s.t $Tv = u$

$\Rightarrow v \in T^{-1}(U)$

$\therefore \forall\ u \in U \cap \text{range } T$, $\exists$ some $v \in T^{-1}(U)$ s.t

$S(v) = u$ □

(b) prove $\text{null } S = \text{null } T$.

$$\text{null } S = \left\{ v \in T^{-1}(\mathcal{U}) : Tv = 0 \right\}$$

$$\text{null } T = \left\{ v \in V : Tv = 0 \right\}$$

$\mathcal{U}$ is a subspace $\therefore 0 \in \mathcal{U}$.

$v \in T^{-1}(\mathcal{U}) \Rightarrow v \in V \Rightarrow \text{null } S \subseteq \text{null } T$

$v \in V,\ 0 = T(v) \in \mathcal{U} \Rightarrow v \in T^{-1}(\mathcal{U}) \Rightarrow \text{null } T \subseteq \text{null } S$.

$\therefore \text{null } S = \text{null } T$.

(*) Prove

$$\dim T^{-1}(U) = \dim \operatorname{null} T + \dim (U \cap \operatorname{range} T)$$

because S is surjective, $\dim U \cap \operatorname{range} T = \dim \operatorname{range} S$

By rank nullity we have

$$\dim \operatorname{domain} S = \dim \operatorname{null} S + \dim \operatorname{range} S.$$

$$\Rightarrow \dim T^{-1}(U) = \dim \operatorname{null} T + \dim (U \cap \operatorname{range} T) \quad \square$$

d) Let $D: P_4(\mathbb{R}) \to P_3(\mathbb{R})$ be differentiation, and let

$$U = \operatorname{Span}(1, x^2) \subseteq P_3(\mathbb{R})$$

$$\iff a_3 = 0$$

determine $\dim D^{-1}(U)$.

by (c): $\dim D^{-1}(U) = \dim \operatorname{null} D + \dim (U \cap \operatorname{range} D)$

$$\forall \; p \in P_3(\mathbb{R}) = a_3 x^3 + a_2 x^2 + a_1 x_1 + a_0$$

$$\exists q = \frac{a_3 x^4}{4} + \frac{a_2 x^3}{3} + \frac{a_1 x^2}{2} + a_0 x + C \in P_4(\mathbb{R})$$

s.t $D(q) = p$. by construction.

$\therefore$ D surjective. and $\operatorname{range} D = P_3(\mathbb{R})$.

$\Rightarrow$ $U \cap \operatorname{range} D = U = \operatorname{span}(1, x^2)$

because $1, x^2$ LI, it is a basis for $U \cap \operatorname{range} D$

and $\dim U \cap \operatorname{range} D = 2$.

$$\operatorname{null} D = \{ p \in P_4(\mathbb{R}) : p' = 0 \} = \mathbb{R}$$

$\therefore$ $\dim \operatorname{null} D = \dim \mathbb{R} = 1$. and $\dim D^{-1} = 1 + 2 = 3$

Basis of D^{-1} : $\text{span}(1, x^2) \to P_4(\mathbb{R})$.

$(1, x, x^3)$ But why?

These polynomials are all LI, and they exist in D^{-1}.

Then by the extension Lemma we can extend this to a basis. But since there are 3 elmts and $\dim D^{-1} = 3$, this must be a basis already. □.